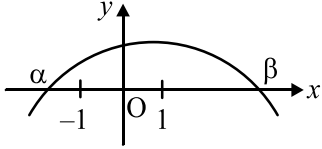




Prayas JEE 2.0 (2024)

Quadratic Equations

DPP-03

- The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is:
 (1) 3 (2) 2
 (3) 4 (4) 5
- The least positive value of 'a' for which the equation, $2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is ____.
- If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, where $ac \neq 0$, then $P(x).Q(x) = 0$ has at least
 (1) Four real roots
 (2) Two real roots
 (3) Four imaginary roots
 (4) None of these
- If the roots of the equations $px^2 + 2qx + r = 0$ and $qx^2 - 2\sqrt{pr}x + q = 0$ be real, then
 (1) $p = q$ (2) $q^2 = pr$
 (3) $p^2 = qr$ (4) $r^2 = pq$
- If $a < b < c < d$, then the roots of the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are
 (1) Real and distinct (2) Real and equal
 (3) Imaginary (4) None of these
- If the roots of the equation $qx^2 + px + q = 0$ where p, q are real, be complex, then the roots of the equation $x^2 - 4qx + p^2 = 0$ are
 (1) Real and unequal (2) Real and equal
 (3) Imaginary (4) None of these
- If a, b, c are integers and $b^2 = 4(ac + 5d^2)$, $d \in N$, then roots of the quadratic equation $ax^2 + bx + c = 0$ are
 (1) Irrational
 (2) Rational & different
 (3) Complex conjugate
 (4) Rational & equal
- The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below.

 Which of the following are correct?
 (1) $\frac{c}{a} < -1$
 (2) $|\beta - \alpha| > 2$
 (3) $f(x) > 0 \forall x \in (0, \beta)$
 (4) $abc < 0$
- The sum of all integral values of $k(k \neq 0)$ for which the equation $\frac{2}{x-1} - \frac{1}{x-2} = \frac{2}{k}$ in x has no real roots, is
- If $ax^2 + (b - c)x + a - b - c = 0$ has unequal real roots for all $c \in R$, then **[Challenger]**
 (1) $b < 0 < a$ (2) $a < 0 < b$
 (3) $b < a < 0$ (4) $b > a > 0$



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (8)
3. (2)
4. (2)
5. (1)

6. (1)
7. (1)
8. (1, 2, 3, 4)
9. (66)
10. (3, 4)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>